

# Worked answers

SOHCAHTOA practice | Suggested marking for original questions | Total: 21 marks

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M = method mark; A = accuracy mark. Accept equivalent correct methods. Accuracy marks normally depend on the relevant method. Correct answers with sufficient working receive full credit.

## 1 Find the opposite side

2 marks

$$\sin 30^\circ = x / 10, \text{ so } x = 10 \times \sin 30^\circ$$

M1

$$x = 5 \text{ cm}$$

A1

The 10 cm side is the hypotenuse. The required side is opposite  $30^\circ$ .

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## 2 Choose the ratio in a rotated triangle

2 marks

$$\cos 42^\circ = y / 13, \text{ so } y = 13 \times \cos 42^\circ$$

M1

$$y = 9.660882... = 9.7 \text{ cm (1 d.p.)}$$

A1

Here y is adjacent to  $42^\circ$ . Turning the triangle does not change the side roles.

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## 3 Find a hypotenuse in context

3 marks

$$\sin 12^\circ = 1.8 / L$$

M1

$$L \sin 12^\circ = 1.8, \text{ so } L = 1.8 / \sin 12^\circ$$

M1

$$L = 8.657521... = 8.7 \text{ m (1 d.p.)}$$

A1

The ramp is the hypotenuse. Multiplying 1.8 by  $\sin 12^\circ$  gives a length smaller than the rise, which cannot be the ramp length.

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## 4 Work backwards to an angle

3 marks

$$\tan \theta = 9 / 14$$

M1

$$\theta = \tan^{-1}(9 / 14)$$

M1

$$\theta = 32.735226... = 32.7^\circ \text{ (1 d.p.)}$$

A1

9/14 is a side ratio, not the angle. Use inverse tangent in degree mode.

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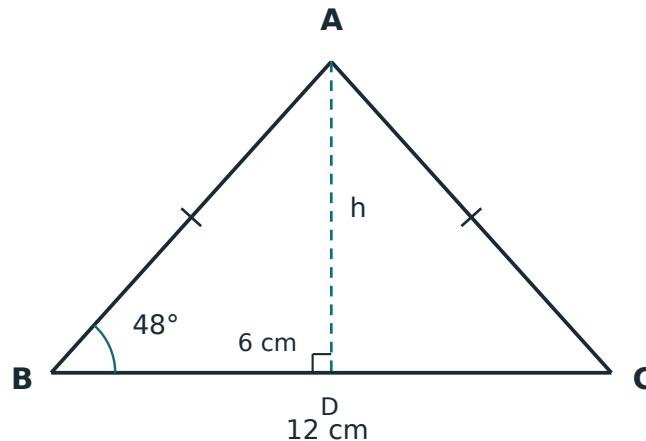
# Worked answers

Question 5 | A hidden right-angled triangle

## 5 Find an area using a perpendicular height

5 marks

Draw the perpendicular AD from A to BC. In an isosceles triangle this splits the base into two equal parts, so  $BD = DC = 6$  cm.



Draw/use AD perpendicular to BC, with  $BD = 6$  cm.

M1

$$\tan 48^\circ = h / 6$$

M1

$$h = 6 \times \tan 48^\circ = 6.663675\dots \text{ cm}$$

A1

$$\text{Area} = \frac{1}{2} \times 12 \times h$$

M1

$$\text{Area} = 39.982050\dots = 40.0 \text{ cm}^2 \text{ (1 d.p.)}$$

A1

Why is the base halved? The two right-angled triangles have equal hypotenuses AB and AC, and share AD. They are congruent, so  $BD = DC$ .

Watch for: using 12 cm as the adjacent side in  $\tan 48^\circ$ . Each right-angled triangle contains only half of the original base. The area formula uses the full base, 12 cm.

Marking note: the base-halving step may be shown in a diagram or in the calculation. Accept a valid equivalent method. The final answer should retain the zero in 40.0 because 1 decimal place was requested.

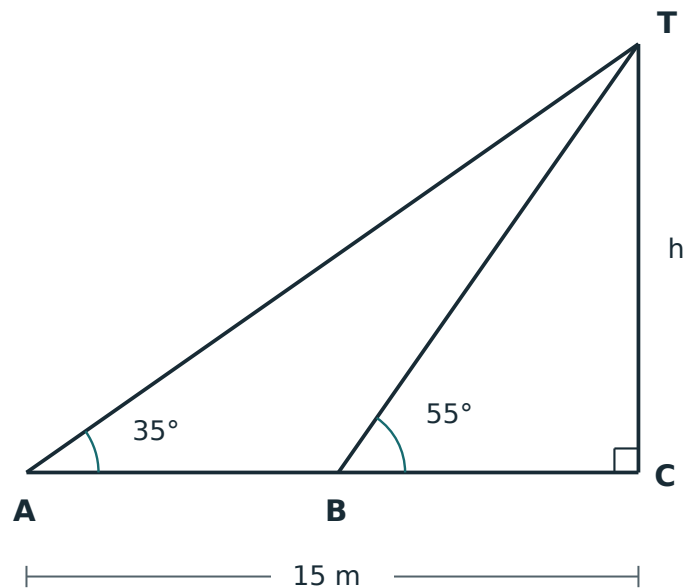
# Worked answers

Question 6 | Two connected right-angled triangles

## 6 Find the distance between two ground points

6 marks

Let the tower height TC be  $h$  metres. First use the larger triangle ATC, then use the same height in triangle BTC.



In triangle ATC:  $\tan 35^\circ = h / 15$

M1

$h = 15 \times \tan 35^\circ = 10.503113... \text{ m}$

A1

In triangle BTC:  $\tan 55^\circ = h / BC$

M1

$BC = h / \tan 55^\circ = 7.354358... \text{ m}$

A1

$AB = AC - BC = 15 - BC$

M1

$AB = 7.645641... = 7.6 \text{ m (1 d.p.)}$

A1

Keep  $h$  and  $BC$  unrounded until the final step. The question asks for  $AB$ , not the height of the tower or the distance  $BC$ .

Check:  $B$  is closer to the tower and has the larger angle,  $55^\circ$ . Both  $BC$  and  $AB$  must be less than  $AC = 15 \text{ m}$ .

Valid alternative: find  $AT = 15 / \cos 35^\circ$ , then use the sine rule in triangle  $ABT$ , whose angles are  $35^\circ$ ,  $125^\circ$  and  $20^\circ$ . This gives the same  $AB$ . Award equivalent method and accuracy credit.